EXERCISE 1:

1)Sum of two matrices

```
#include<stdio.h>
int main()
{
    int a[10][10], b[10][10], c[10][10];
    int i,j,r,c1;

   printf("Enter rows and columns: ");
    scanf("%d%d",&r,&c1);

    printf("Enter first matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c1;j++)
            scanf("%d",&a[i][j]);

    printf("Enter second matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c1;j++)
            scanf("%d",&b[i][j]);

    printf("Sum Matrix:\n");
    for(i=0;i<r;i++)
    {
        for(j=0;j<c1;j++)
        {
            c[i][j]=a[i][j]+b[i][j];
            printf("%d ",c[i][j]);
```

```
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

Enter rows and columns: 2 2

Enter first matrix:

1 2

3 4

Enter second matrix:

5 6

7 8

Sum Matrix:

6 8

10 12

2)Write a C program to read n number of values in an array and display them in reverse order

```
#include<stdio.h>
int main()
{
    int a[100],n,i;

    printf("Enter number of elements: ");
    scanf("%d",&n);

    printf("Enter elements:\n");
    for(i=0;i<n;i++)
```

```
        scanf("%d",&a[i]);

    printf("Reverse order:\n");
    for(i=n-1;i>=0;i--)
        printf("%d ",a[i]);

    return 0;
}
```

Output:

Enter number of elements: 3

Enter elements:

50 60 70

Reverse order:

70 60 50

3)Count the total number of duplicate elements in an array

```
#include<stdio.h>
int main()
{
    int a[100],n,i,j,count=0;

    printf("Enter number of elements: ");
    scanf("%d",&n);

    printf("Enter elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    for(i=0;i<n;i++)
```

```
   {
      for(j=i+1;j<n;j++)
      {
         if(a[i]==a[j])
         {
            count++;
            break;
         }
      }
   }

   printf("Total duplicate elements = %d",count);

   return 0;
}
```

Output:

Enter number of elements: 5

Enter elements:

30 40 50 60 50

Total duplicate elements = 1

4)Print unique element in an array

```
#include<stdio.h>
int main()
{
   int a[100],n,i,j,flag;

   printf("Enter number of elements: ");
   scanf("%d",&n);
```

```
    printf("Enter elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Unique elements are:\n");

    for(i=0;i<n;i++)
    {
        flag=0;
        for(j=0;j<n;j++)
        {
            if(i!=j && a[i]==a[j])
            {
                flag=1;
                break;
            }
        }
        if(flag==0)
            printf("%d ",a[i]);
    }

    return 0;
}
```

Output:

Enter number of elements: 7

Enter elements:

10 20 30 20 10 40 15

Unique elements are:

30 40 15

5)Insert in sorted array

```
#include<stdio.h>
int main()
{
   int a[100],n,i,pos,value;

   printf("Enter number of elements: ");
   scanf("%d",&n);

   printf("Enter sorted elements:\n");
   for(i=0;i<n;i++)
      scanf("%d",&a[i]);

   printf("Enter value to insert: ");
   scanf("%d",&value);

   for(i=0;i<n;i++)
   {
      if(value<a[i])
      {
         pos=i;
         break;
      }
   }

   for(i=n;i>pos;i--)
```

```
        a[i]=a[i-1];

    a[pos]=value;

    printf("Array after insertion:\n");
    for(i=0;i<=n;i++)
        printf("%d ",a[i]);

    return 0;
}
```

Output:

Enter number of elements: 5

Enter sorted elements:

20 25 30 35 40

Enter value to insert: 26

Array after insertion:

20 25 26 30 35 40

6)Insert in unsorted array

```
#include<stdio.h>

int main()
{
    int a[100], n, i, pos, value;

    printf("Enter number of elements: ");
    scanf("%d",&n);

    printf("Enter elements:\n");
```

```
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Enter position: ");
    scanf("%d",&pos);

    printf("Enter value: ");
    scanf("%d",&value);

    for(i=n;i>=pos;i--)
        a[i]=a[i-1];

    a[pos-1]=value;

    printf("Array after insertion:\n");
    for(i=0;i<=n;i++)
        printf("%d ",a[i]);

    return 0;
}
```

Output:

Enter number of elements: 5

Enter sorted elements:

20 25 30 35 40

Enter value to insert: 26

Array after insertion:

20 25 26 30 35 40

7)Matrix multiplication

```c
#include<stdio.h>

int main()
{
   int a[10][10], b[10][10], c[10][10];
   int i,j,k,r1,c1,r2,c2;

   printf("Enter rows and columns of first matrix: ");
   scanf("%d%d",&r1,&c1);

   printf("Enter rows and columns of second matrix: ");
   scanf("%d%d",&r2,&c2);

   if(c1!=r2)
   {
      printf("Matrix multiplication not possible");
      return 0;
   }

   printf("Enter first matrix:\n");
   for(i=0;i<r1;i++)
      for(j=0;j<c1;j++)
         scanf("%d",&a[i][j]);

   printf("Enter second matrix:\n");
   for(i=0;i<r2;i++)
      for(j=0;j<c2;j++)
         scanf("%d",&b[i][j]);
```

```
    for(i=0;i<r1;i++)
    {
        for(j=0;j<c2;j++)
        {
            c[i][j]=0;
            for(k=0;k<c1;k++)
                c[i][j]+=a[i][k]*b[k][j];
        }
    }

    printf("Product Matrix:\n");
    for(i=0;i<r1;i++)
    {
        for(j=0;j<c2;j++)
            printf("%d ",c[i][j]);
        printf("\n");
    }

    return 0;
}
```

Output:

Enter rows and columns of first matrix: 3 3

Enter rows and columns of second matrix: 3 3

Enter first matrix:

1 2 3

4 5 6

7 8 9

Enter second matrix:

3 5 6

7 9 1

8 0 5

Product Matrix:

41 23 23

95 65 59

149 107 95

8)Matrix transpose

```
#include<stdio.h>

int main()
{
   int a[10][10], i, j, r, c;

   printf("Enter rows and columns: ");
   scanf("%d%d",&r,&c);

   printf("Enter matrix:\n");
   for(i=0;i<r;i++)
      for(j=0;j<c;j++)
         scanf("%d",&a[i][j]);

   printf("Transpose Matrix:\n");
   for(i=0;i<c;i++)
   {
      for(j=0;j<r;j++)
         printf("%d ",a[j][i]);
```

```
        printf("\n");
    }

    return 0;
}
```

Output:

Enter rows and columns of first matrix: 3 3

Enter rows and columns of second matrix: 3 3

Enter first matrix:

1 2 3

4 5 6

7 8 9

Enter second matrix:

3 5 6

7 9 1

8 0 5

Product Matrix:

41 23 23

95 65 59

149 107 95

9)Right diagonal sum

```
#include<stdio.h>

int main()
{
    int a[10][10], n, i, j, sum=0;

    printf("Enter order of matrix: ");
```

```
    scanf("%d",&n);

    printf("Enter matrix:\n");
    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            scanf("%d",&a[i][j]);

    for(i=0;i<n;i++)
        sum+=a[i][n-1-i];

Check

    return 0;
}
```

Output:

Enter order of matrix: 3

Enter matrix:

1 2 3

4 5 6

7 8 9

Right diagonal sum = 15

10)Check for identity matrix

```
#include<stdio.h>

int main()
{
    int a[10][10], n, i, j, flag=1;
```

```
    printf("Enter order of matrix: ");
    scanf("%d",&n);

    printf("Enter matrix:\n");
    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            scanf("%d",&a[i][j]);

    for(i=0;i<n;i++)
    {
        for(j=0;j<n;j++)
        {
            if(i==j && a[i][j]!=1)
                flag=0;
            if(i!=j && a[i][j]!=0)
                flag=0;
        }
    }

    if(flag)
        printf("Identity Matrix");
    else
        printf("Not an Identity Matrix");

    return 0;
}
```

Output:

Enter order of matrix: 2

Enter matrix:

1 0

0 1

Identity Matrix

11)To find no.of vowels, consonants, digits and white spaces

```
#include<stdio.h>
#include<string.h>

int main()
{
    char str[100];
    int i,v=0,c=0,d=0,s=0;

    printf("Enter a string: ");
    fgets(str,100,stdin);

    for(i=0;str[i]!='\0';i++)
    {
if(str[i]=='a'||str[i]=='e'||str[i]=='i'||str[i]=='o'||str[i]=='u'||str[i]=='A'||str[i]=
='E'||str[i]=='I'||str[i]=='O'||str[i]=='U')
            v++;
        else if((str[i]>='A'&&str[i]<='Z')||(str[i]>='a'&&str[i]<='z'))
            c++;
        else if(str[i]>='0'&&str[i]<='9')
            d++;
        else if(str[i]==' ')
            s++;
```

```
    }

    printf("Vowels=%d\nConsonants=%d\nDigits=%d\nSpaces=%d",v,c,d,s);

    return 0;
}
```

Output:

Enter a string: Hello World

Vowels=3

Consonants=7

Digits=0

Spaces=1

12)Frequency of a character in a string

```
#include<stdio.h>

int main()
{
    char str[100], ch;
    int i,count=0;

    printf("Enter string: ");
    fgets(str,100,stdin);

    printf("Enter character: ");
    scanf("%c",&ch);

    for(i=0;str[i]!='\0';i++)
        if(str[i]==ch)
```

```
        count++;

    printf("Frequency = %d",count);

    return 0;
}
```

Output:

Enter a string: Hello World

Vowels=3

Consonants=7

Digits=0

Spaces=1

13)Count words in a string

```
#include<stdio.h>

int main()
{
    char str[100];
    int i,words=1;

    printf("Enter sentence: ");
    fgets(str,100,stdin);

    for(i=0;str[i]!='\0';i++)
        if(str[i]==' ')
            words++;

    printf("Number of words = %d",words);
```

```
    return 0;
}
```

Output:

Enter sentence: Data Structure

Number of words = 2

14)Sort string array

```
#include<stdio.h>
#include<string.h>

int main()
{
    char str[5][20], temp[20];
    int i,j;

    printf("Enter 5 strings:\n");

    for(i=0;i<5;i++)
        scanf("%s",str[i]);

    for(i=0;i<4;i++)
    {
        for(j=i+1;j<5;j++)
        {
            if(strcmp(str[i],str[j])>0)
            {
                strcpy(temp,str[i]);
                strcpy(str[i],str[j]);
```

```
            strcpy(str[j],temp);
          }
        }
    }

    printf("Sorted strings:\n");

    for(i=0;i<5;i++)
        printf("%s\n",str[i]);

    return 0;
}
```

Output:

Enter 5 strings:

Kavi

Anu

Rohi

Pradee

Prani

Sorted strings:

Anu

Kavi

Pradee

Prani

Rohi

15)Toggle case of a sentence (lowercase to uppercase and vice versa)

```
#include<stdio.h>
```

```
int main()
{
    char str[100];
    int i;

    printf("Enter a sentence: ");
    fgets(str,100,stdin);

    for(i=0;str[i]!='\0';i++)
    {
        if(str[i]>='A'&&str[i]<='Z')
            str[i]=str[i]+32;
        else if(str[i]>='a'&&str[i]<='z')
            str[i]=str[i]-32;
    }

    printf("After Toggle Case:\n%s",str);

    return 0;
}
```

Output:

Enter a sentence: My name is Kavi

After Toggle Case:

mY NAME IS kAVI